CANDY
DIALOGUE ENGINE

Basic Dialogue

1.1.0

This guide describes how to create a basic dialogue in the Candy Dialogue Creator, and test it in Godot.

If you haven't done so already, follow the Basic Setup guide.

0. Preparation

This tutorial will require some portrait files. You can download some from our GitHub:

1. Follow the link above and download the "Portraits" folder and its contents.
2. Merge the Portraits folder it with your Candy_DE/Media/Characters/Portraits folder.
3. Open your project in the Godot editor to import the portraits (if it's already opened, switch to the Godot editor window).

1. Dialogue Writing

Candy DE Basics

You can think of Candy DE as having three layers: the dialogues, which you write; the engine, which reads dialogue lines; and various UI elements, which display or play dialogue text and media to the player.

The engine processes dialogues one line at a time, and each line acts as an instruction that the engine follows: display text on screen, play audio, modify a variable, display player choices, etc.

Depending on the line, the engine might defer to UI elements. For example, when it processes a spoken line (a character speaking), it does a bunch of internal operations to prepare the line for display, then calls upon the dialogue box to actually display the line on screen; when the engine processes a media command, it calls upon a media player (e.g. AudioStreamPlayer) to play the media; and so on.

It's a very simple design:

Dialogues → Engine → UI Elements

Dialogue Basics

Structure

Dialogues are built hierarchically: Conversations contain Blocks, and Blocks contain Lines.

You can write all of your game's dialogues in a single dialogue file, or you can break dialogues into multiple files if you find this more convenient. To illustrate:

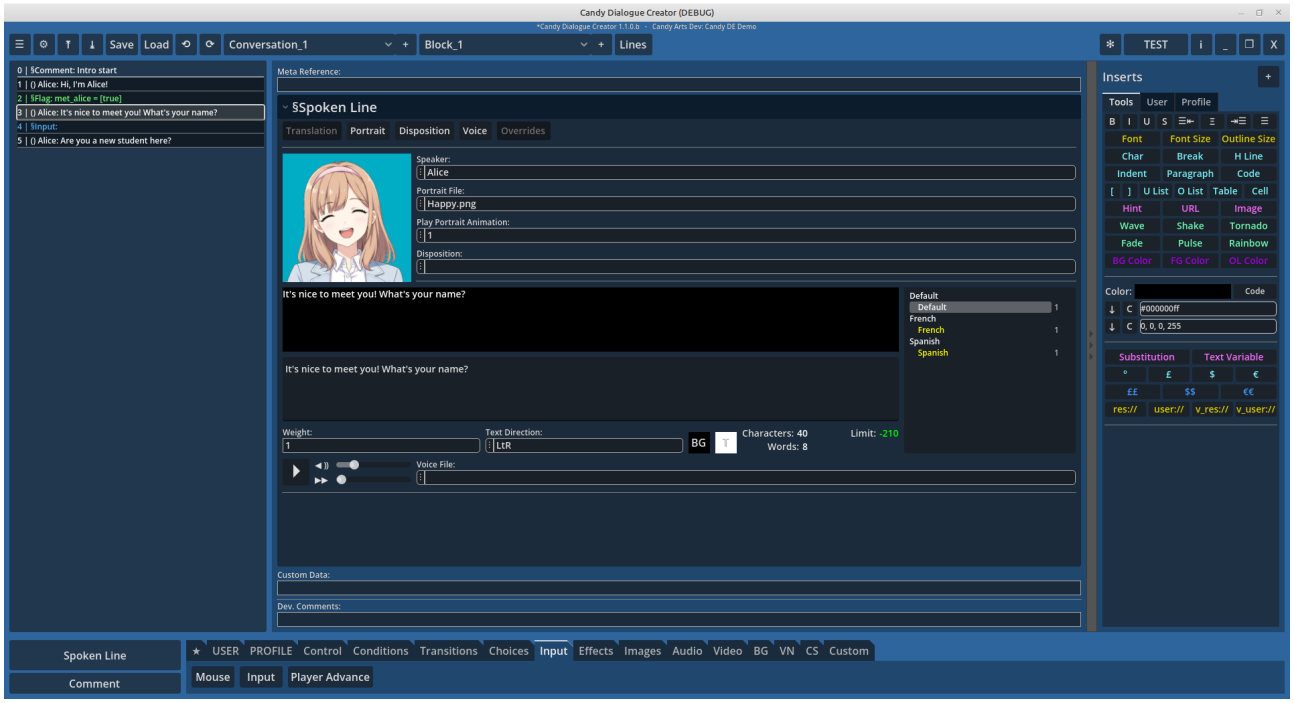
Dialogue Files → Conversations → Blocks → Lines

There are no hard rules about splitting your dialogues into Conversations or Blocks. Generally, you would treat each Conversation as a single scene, while Blocks would be used to strategically separate parts of a scene.

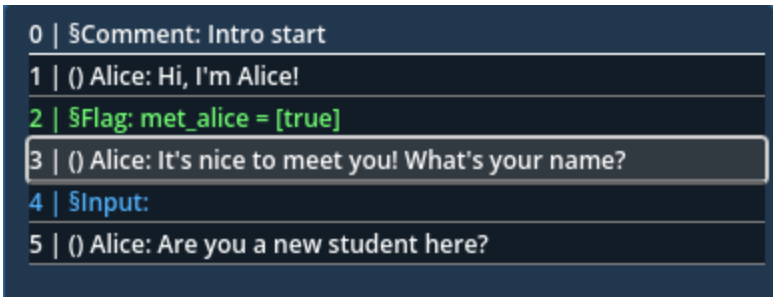
Lines

Two types of lines exist: spoken lines displayed on screen as character speech, and command lines which have various effects on dialogues or on your game, such as modifying variables or calling functions, transitioning to other parts of a dialogue, condition branching, player choices, or playing media.

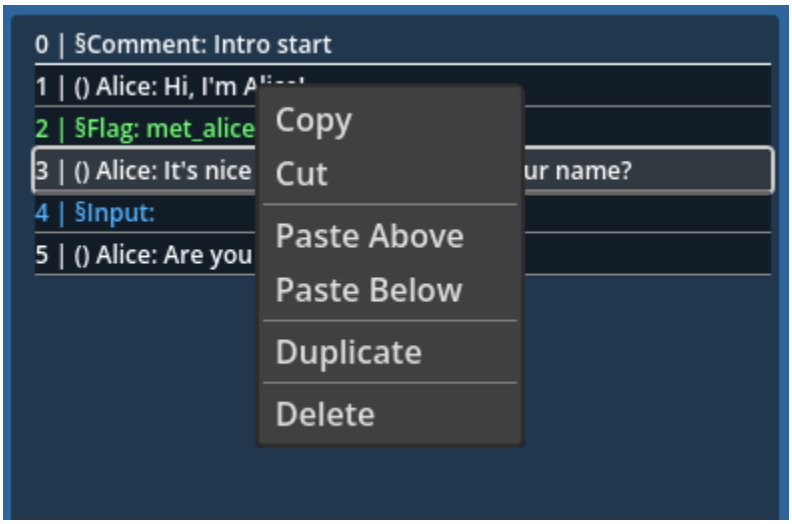
Editor UI Basics



The area to the left displays all the lines inside the selected dialogue Block. An empty Comment line is automatically created in any new Block.



You can left-click the name of a line to select it, or you can right-click it to copy, cut, paste, duplicate or delete a line:



You can re-order lines by clicking and dragging them. Use SHIFT + Click or CTRL + Click to select multiple lines at once. Only the last selected line will display data in the central area.

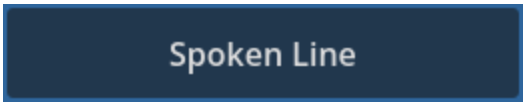
You can navigate Conversations and Blocks at the top of the screen:

Lines are added by clicking the buttons in the bottom area of the screen. Commands are sorted into categories:

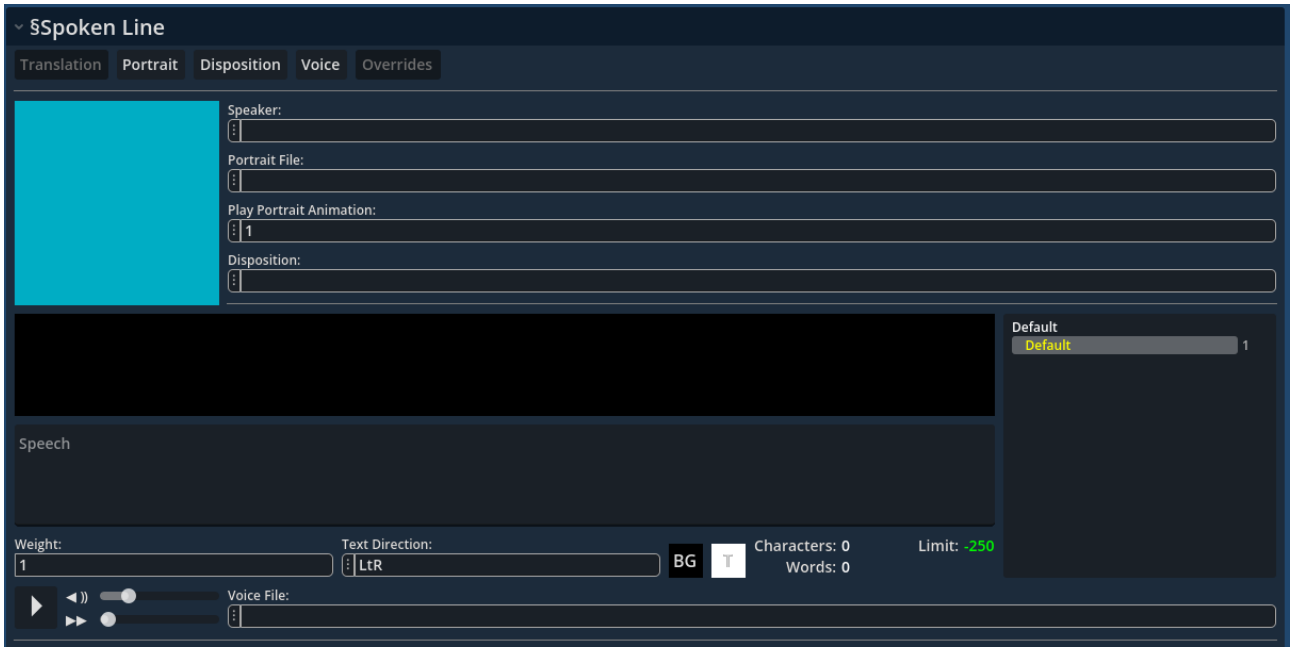
2. Example Dialogue

Spoken Lines

Start by clicking the large "Spoken Line" button in the bottom-left corner. This will add a new spoken line to the current Block.



The line's data will display in the central area of the screen. The data fields will be empty:



Speaker, Portrait and Speech

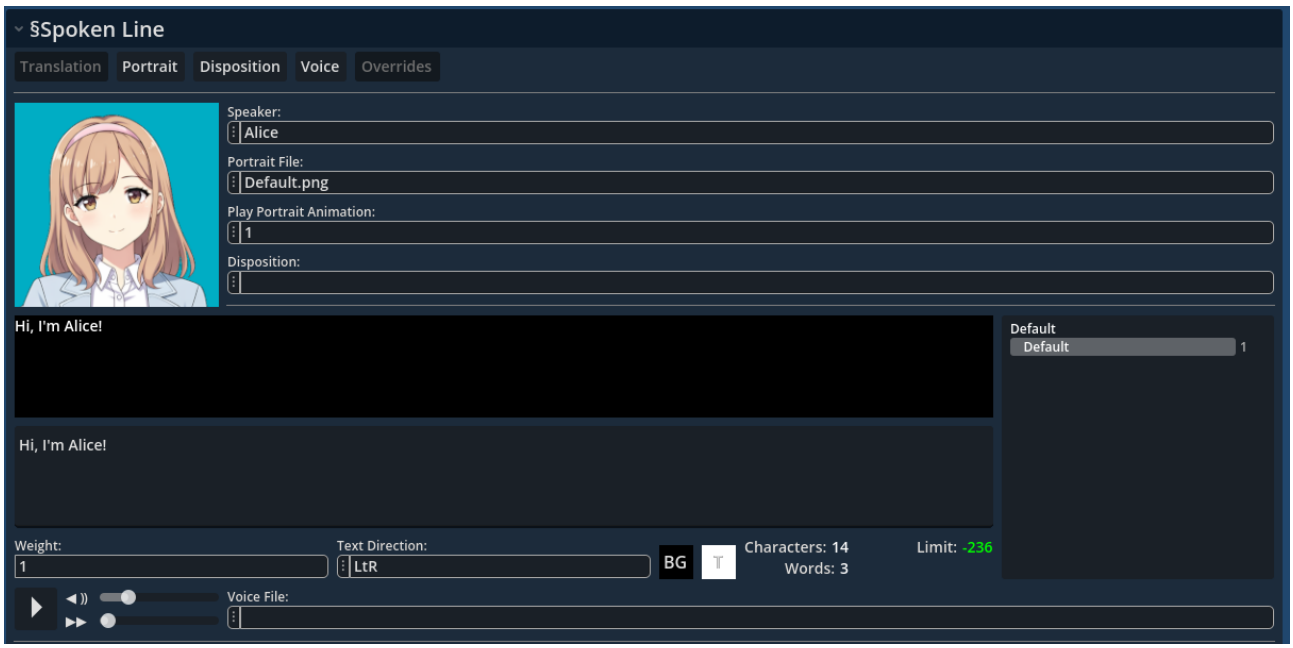
In the "Speaker" field, write "Alice" (without the quotation marks) or click the (⋮) button to the right and select "Alice" from the list.

In the "Portrait File" field, click the (⋮) button to the right and select a portrait (e.g. "Default.png"). Notice that you can see a preview of each portrait by hovering them in the list. Once you select a portrait, it will be displayed to the left.

Click the portrait area: a color picker menu appears. Select a background color of your choice for the portrait. This color only appears in Candy Dialogue Creator, it has no effect in Candy Dialogue Engine or your game.

In the "Speech" field, write "Hello, I'm Alice.".

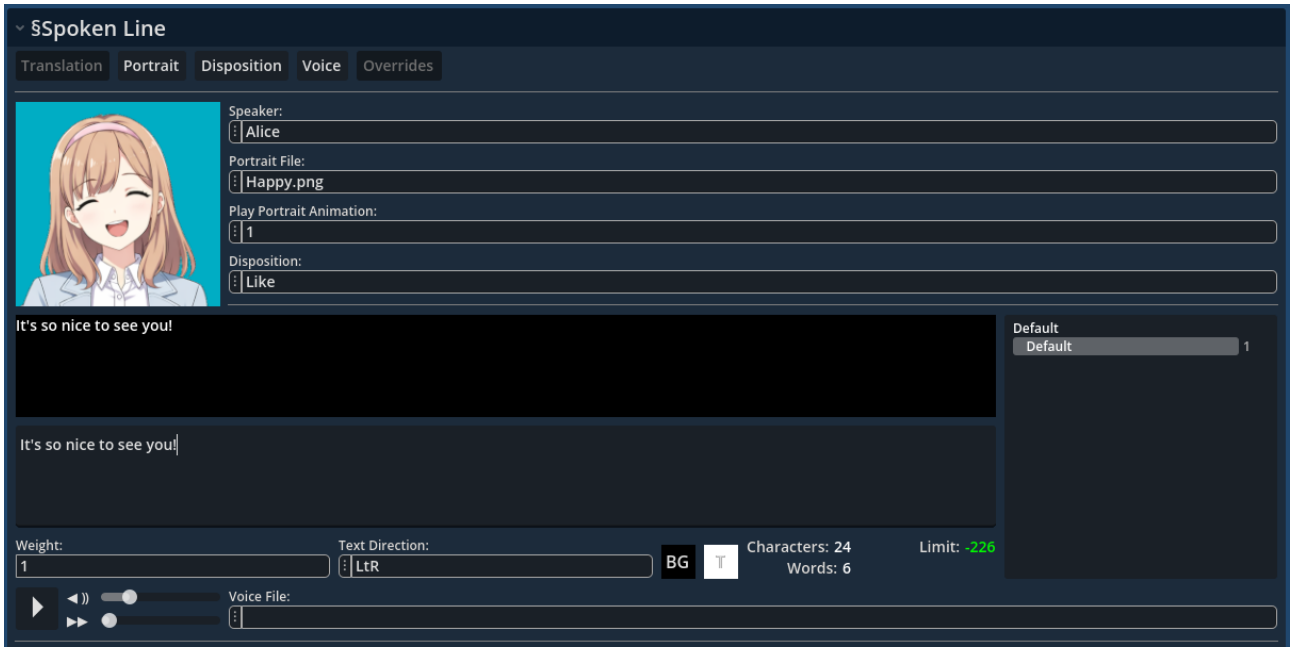
That's it, you've just written your first dialogue line!



Dispositions

Now let's make Alice say the line only if she likes the player: in the "Disposition" field, write "Like" (without the quotation marks) or click the (⋮) button and select "Like" from the list if it's present (it only appears if you've imported or added it in your settings).

For Alice's speech, write "It's so nice to see you!" and select the "Happy.png" portrait.



Dispositions are an easy and quick way to make certain spoken lines conditional on the speaker's mood or disposition towards the player. Simply write a disposition (e.g. "Like") in the "Dispositions" field and, if the speaker's disposition matches in the game when the line is

processed, the line will be displayed. If it's not a match, the line will be skipped.

If you hover the Disposition field long enough with the mouse, a tooltip will appear with a lot of additional information.

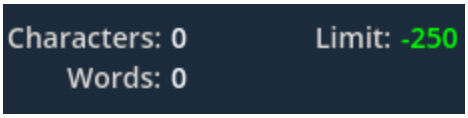
Don't worry about it for now, this is advanced stuff that you don't need to know about for basic dialogues. In fact, even advanced dialogues won't use all of these options. Like most Candy DE features, it's there if you need it, and it can be ignored entirely if you don't have a use for it.

Character Limit & Stats

Finally, look below the Speech area to see some numbers.

"Characters" and "Words" indicate the number of characters and words you've written in the spoken text.

"Limit" tells you whether you've exceeded the maximum recommended character limit. This is helpful when your dialogue display UI supports a limited number of characters.



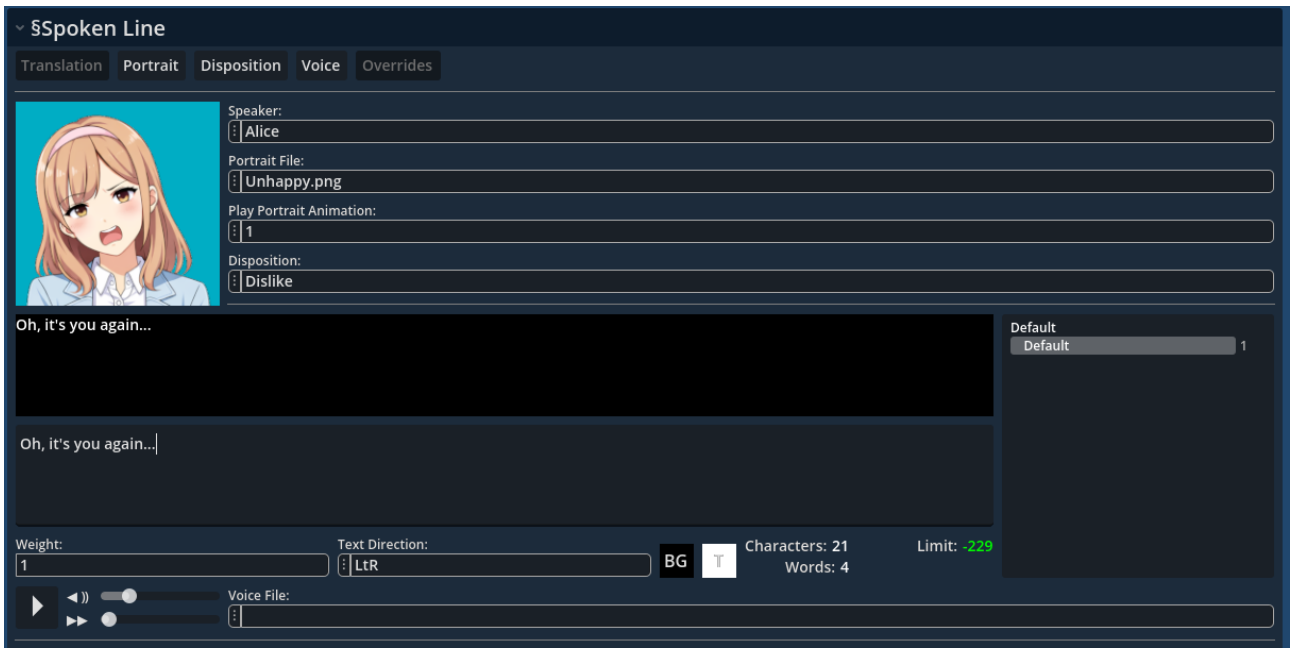
A green number indicates the characters remaining before you reach the limit, a yellow number means you're close to reaching the limit and you should consider splitting the text over another line, and a red number indicates by how many characters you've exceeded the limit.

The numbers are customizable in the Settings Menu. By default, the limit is set to 250 and the number turns yellow when you're 50 characters away from the limit.

Extra Line

To make sure you understood the basics of spoken lines, add a third line for Alice. Once again, choose her as the speaker, make her say "Oh, it's you again...", and choose the "Unhappy" portrait.

In the Disposition field, write or select "Dislike". This will demonstrate how dispositions work when we test the dialogue later: the first line (with disposition "Like") will show, but the second line (disposition "Dislike") won't show because Alice's default disposition is set to "Like" in Candy DE's database.



Command Lines

Let's now learn about command lines: look at the menu on the left of the screen with the command line buttons. Command lines might seem technical, but they're very easy to use once you understand what they do.

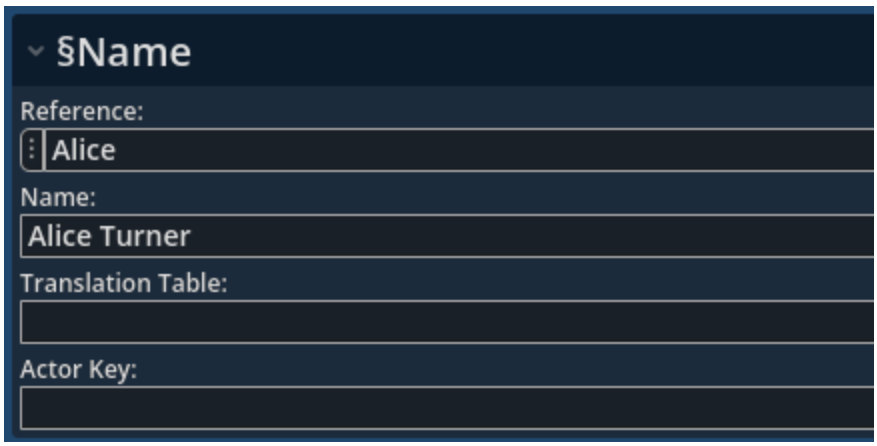
Control commands

Control commands are used for changing variables, calling functions, or emitting signals.

Click the "Name" button to add the §Name command to your dialogue.

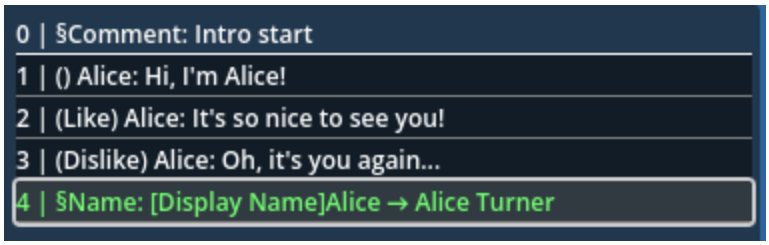
In the code, commands are prefixed with the '§' symbol so the engine can process them differently from Spoken Lines. The documentation (like this guide) follows this convention as well for clarity.

In the §Name command's Reference field, write "Alice" again (without quotation marks) or select Alice from the (⁝) list. In the Value field, write a new name for Alice (e.g. "Alice Turner" or "Alice the Evil Overlord").



The Name command will change how Alice's name is displayed when she speaks. However, it does not change her reference: you will still write "Alice" inside dialogue lines.

Notice how, in the line list to the left, the §Name command line is green, while the Speech line is white? These colors will make it easy to quickly identify lines by category when you browse through your dialogues for review or editing.

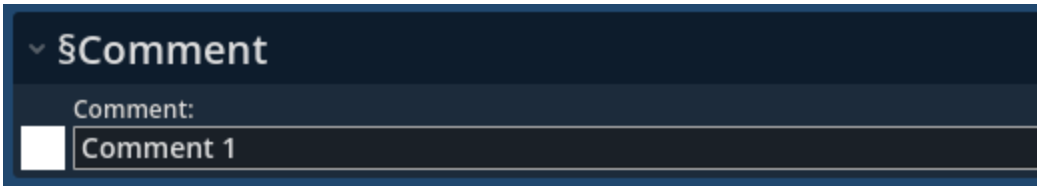


The colors can be customized in the Settings Menu.

Comments

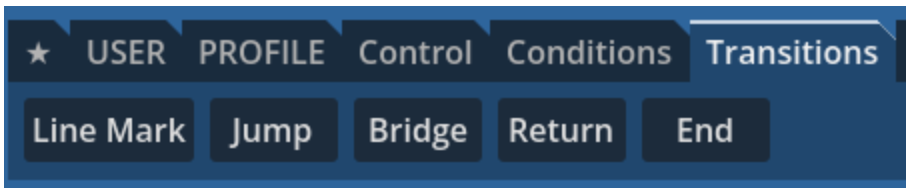
If you'd like to try, feel free to also add a Comment line by clicking the "Comment" button in the bottom-left corner of the screen. Comments are only for helping you write and edit dialogues, they have no effect in your game.

Write whatever you want in the "Comment" field, and change the text color by clicking the white square button, then selecting a color. This is just a small convenience feature.



Transition commands

Transition commands handle dialogue flow: they are used for switching to another Block during dialogue, or to End the dialogue.



There are two options for transitioning forward: §Bridge and §Jump.

§Bridge is a momentary detour in the dialogue: Candy DE will switch to the indicated Block, process its lines, and then come back to the previous Block so it can continue where it left off.

§Jump is a permanent change in the dialogue: Candy DE will not return to the previous Block after processing the new Block.

§Return transitions backward: if a Block was reached through a §Bridge transition, §Return will return to that Block, and continue where it left off (i.e. after the §Bridge command).

§Line Mark acts as a line reference: §Bridge and §Jump can transition to a specific line using line indexes, but indexes are fragile (they'll change if you add or remove lines). Instead, it's much more reliable to target a §Line Mark's reference.

§End just ends the dialogue (more on that later).

Add a §Jump command, as we don't want to return to Block_1, and edit its data like so:

Leave the "Conversation" field empty: Candy DE will automatically default to the same Conversation that the §Jump command is located in.

In the "Block" field, select Block_2 (if you leave the Block field empty, Candy DE also defaults to the same Block the §Jump command is located in).

Leave the Line field empty: it's only used when transitioning to a specific line inside a Block. If you leave it empty, it transitions to line 0 -- the first line of the Block -- which is what we want to do.

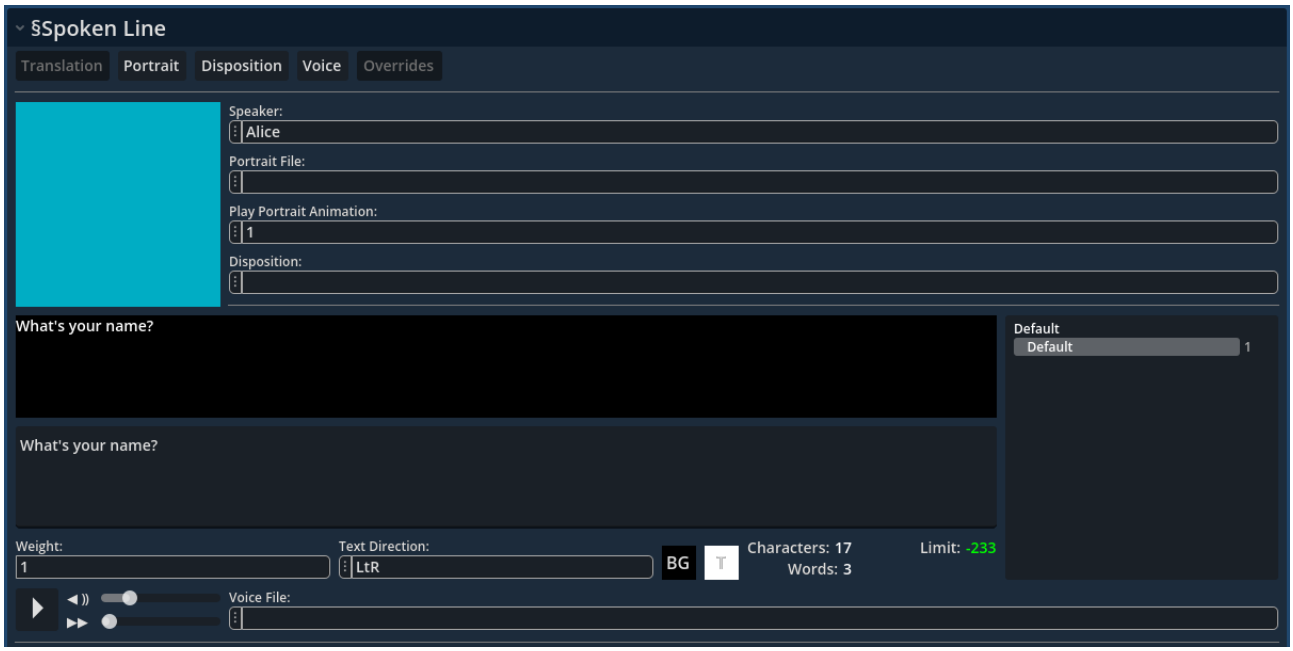
Finally, click the "Go to Block ⇾" button below the data fields: this will automatically create the missing Block_2 inside Conversation_1 and make the UI switch to it. The lines on screen will disappear of course, since we're now in Block_2. If you ever want to create a Block without switching to it immediately, you can CTRL + Click the "Go to Block ⇾" button instead.

To navigate back to Block_1 at any time, you can use the Conversation and Block selector buttons at the very top of the screen.

More dialogue

Just to make sure everything worked when we test the dialogue, add a Spoken Line to Block_2.

Write "Alice" again as the speaker, leave the Portrait field empty to use her default portrait, and write "What's your name?" in the Speech field. Don't write a disposition, this line shouldn't be conditional.



Since we changed Alice's display name with the §Name command earlier in Block_1, when the dialogue runs, her name on screen for this line will be "Alice Turner" (or any name you gave her).

Now, add a second Spoken Line to Block_2.

This time, write "John" in the Speaker field. John doesn't exist in Candy DE's 'actors' dictionary, so his name doesn't show up in the actor list if you click the (⁝) button.

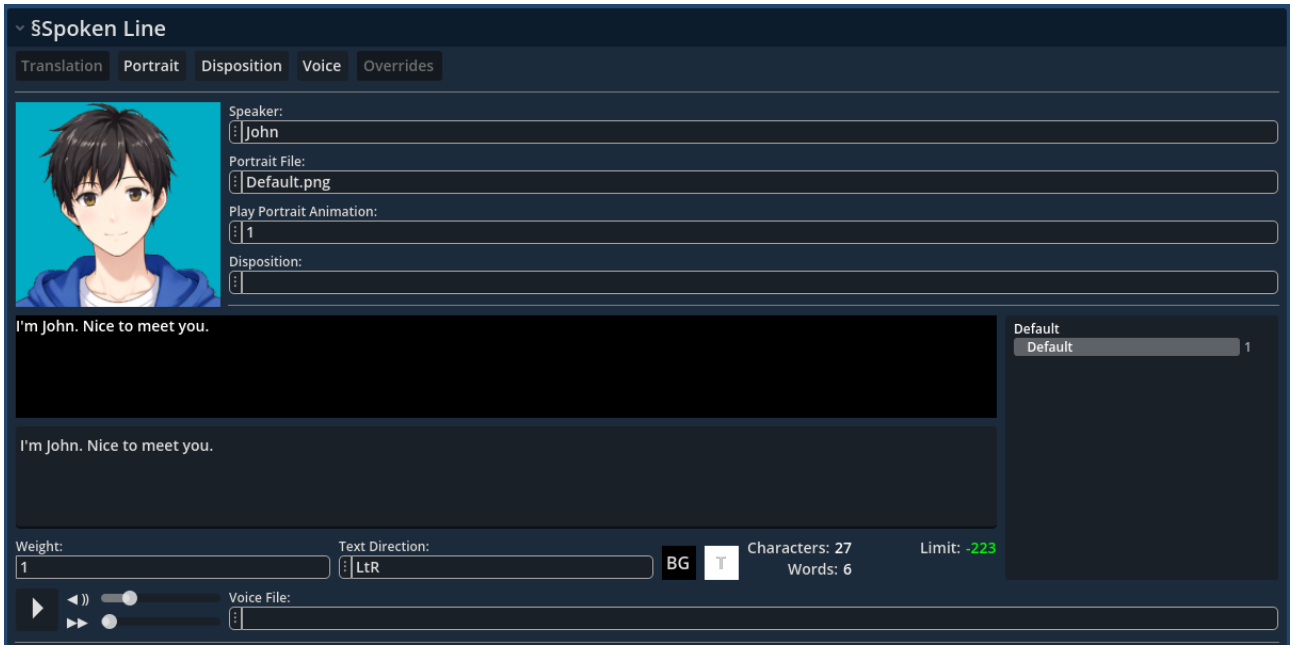
After you have written "John" as the speaker, click the (⁝) button for the "Portraits" field and select "Default.png" from the list.

When an actor doesn't exist in the dictionary, Candy DE will display the reference as the speaker's name for the line. It just won't have custom data for John the way it does for Alice, such as custom

text color, dispositions, and so on. For minor characters who don't need an entry in your game's database, this is fine.

Although John doesn't exist in the actors dictionary, a portrait is available for him. That's because a Portraits folder exists for John in your project's resources.

For John's Speech, write "I'm John. Nice to meet you."



Roles

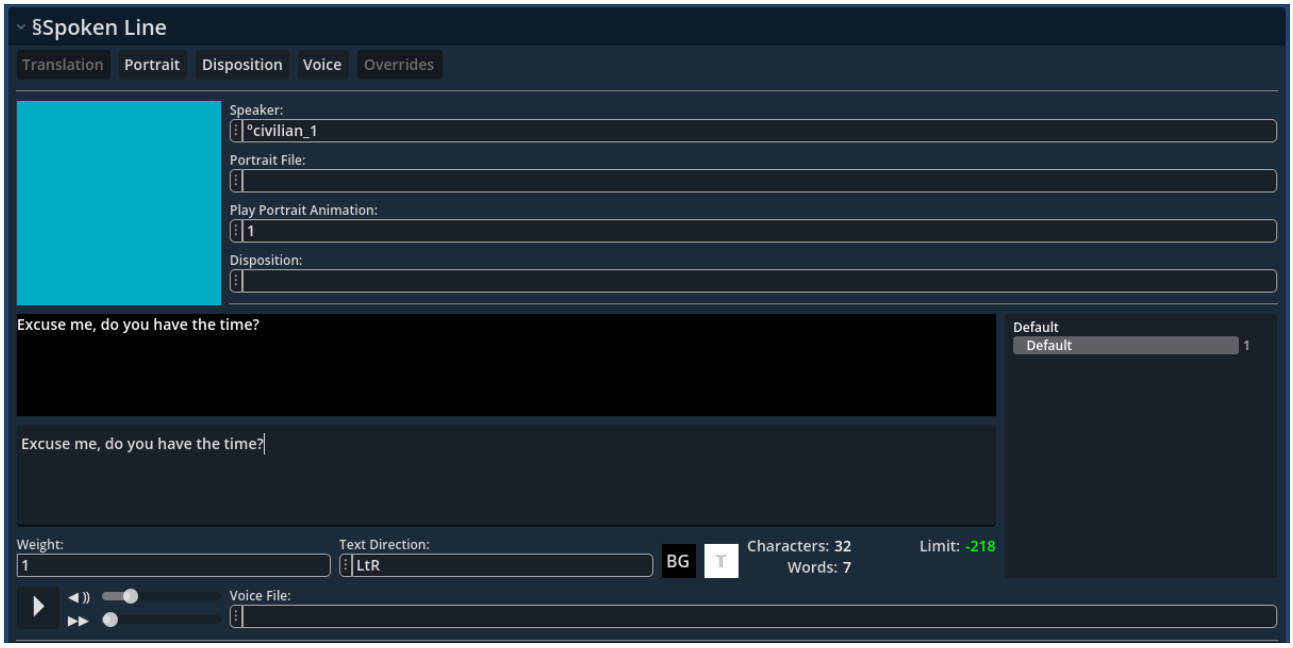
Let's also learn about speaker roles while we're at it:

Add a §Role command, and for the Role field, select "°civilian_1". In the Reference field, select "construction_worker".

Once the §Role command is setup, add a Spoken Line once more. In the Speaker field, click the (⋮) button, hover the "Roles" option, and select "°civilian_1".

Roles always start with the role symbol (default: '°'): this is how Candy DE knows whether it's dealing with a role reference or an actor reference.

In the Speech field, write "Excuse me, do you have the time?"



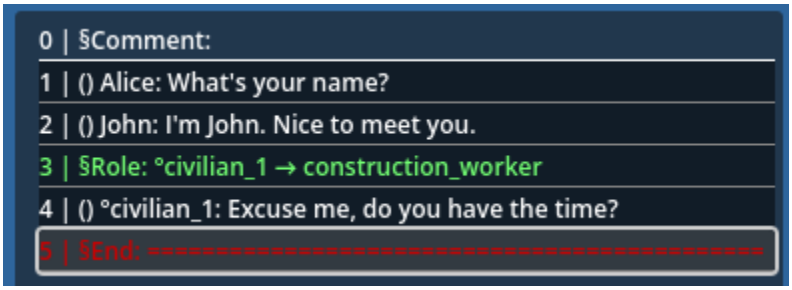
Although construction_worker has portraits and will be assigned to the °civilian_1 role by the earlier §Role command, the portraits won't display in the portrait list since the assignment will occur at runtime.

> You could manually write a portrait name in the Portrait field. Ideally, you would ensure any actors that can potentially be assigned to a role have portraits for the matching name.
>
> If the written portrait doesn't exist for any actor assigned to a role, Candy DE will default to the **Default.png** portrait.

End

Since this is just a brief tutorial, we'll make the dialogue end here.

Add the §End command, in the Transition Commands category:



Notice that it doesn't accept any data. That's because it serves a single purpose: ending dialogue. As soon as the dialogue engine runs this command, dialogue ends. Simple.

In the line list, the §End command is a big red line: this helps it stand out and visually mark where dialogue will stop.

Note that you don't always need to add the §End command. When the dialogue engine reaches the end of a Block, and if that Block was **not** reached through a §Bridge command, then the dialogue will end automatically because there's no lines left to process.

In this case, we reached Block_2 through a §Jump command, so the dialogue would end at the end of Block_2 anyway. That said, it's good practice to always add an §End command: it reduces the chances that you might forget to add it when needed, and it visually shows where dialogue is expected to end while you're editing.

3. Testing Dialogues

Exporting

Now that we have a minimal dialogue, click the "Test" button in the top-right corner of the UI. And just like that, your dialogue has been exported to your project resources!

To make sure it worked, go to your project folder, and from there, navigate to **Candy_DE/Dialogues**. You should see a file named "Test_Dialogue.txt".

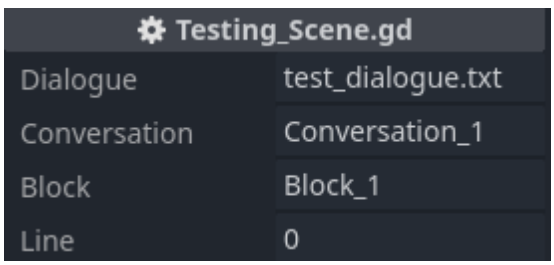
Note that this method is only for quickly exporting a test file. Any previous versions of the **test_dialogue.txt** file will be overwritten each time you click "Test". When your dialogue is complete and you're ready to make a proper export, the Export Menu, inside the Main Menu, offers additional options, including giving custom names to exported files.

Test Scene

Running dialogues

Now let's test your dialogue. Go back to the Godot Editor and open the scene named **Dialogue Test Scene.tscn** found in **Candy_DE/Scenes**.

Run the scene, and if you've followed all instructions correctly, your dialogue should immediately begin. That's it: two-click dialogue testing!



By default, the testing scene is configured to start dialogues at Conversation_1, Block_1.

To start at a different Conversation or Block, select the test scene's root node then, in the Inspector, change the **Conversation**, **Block** and **Line** exported variables to the values you want:

Note that the test scene is empty and only starts dialogue. If you run it directly, you won't see game elements on screen, and some commands may have no effect (or may even crash). It can only display UI elements from the Candy_UI.

See the **Test Scene** guide for more information on how to fully test dialogues.

Success

If you did everything correctly, you'll see the following dialogue, in sequence:

> **Alice: Hello, I'm Alice.**
>
> **Alice Turner: What's your name?**
>
> **John: I'm John. Nice to meet you.**
>
> **Construction Worker: Excuse me, do you have the time?**

Errors

If the dialogue doesn't run:

1. Check that the starting Conversation and Block in the test scene match those in your dialogue.
2. Check that the **test_dialogue.txt** file is located in the **Candy_DE/Dialogues folder**.
3. Check that the **Candy_DE** folder is inside your Godot project's root folder.
4. In Candy Dialogue Creator, go to Main Menu → Settings → Project Files and verify that the Project Folder path is correct (i.e. it must point to your project's root folder).
5. Check that the "Custom Folders" box (to the right of the Project Folder path) is NOT ticked (the paths underneath must be grayed out).

Ending Dialogues

As previously mentioned, dialogues will end on their own when the last line has been processed or when encountering an §End command.

In both cases, the Godot Editor output area will print "Dialogue Ended" to let you know the dialogue ended properly and didn't stop due to an error.

4. Conclusion

You've now setup both Candy DE and Candy DC, and learned the basics of creating dialogues:

- Conversations and Blocks;
- Adding Lines;
- Speaker References (roles and actors);
- Portraits;
- The concept of Dispositions;
- Transitions;
- Control commands such as "Name" and "Role";
- Exporting dialogues for testing;
- Running a test dialogue in the Godot Editor;

You'll probably want to learn about a few of the following intermediate features:

- Alternate dialogue or portrait display modes (visual novels, speech bubbles, etc.);
- Adding actors, roles, or other information to Candy DE's database;
- More info on Dispositions;
- Conditions and branching;
- Player choices;
- Playing/displaying media;
- Calling functions or emitting signals;
- Displaying variables in spoken text;
- Customizing the UI;
- Using the Writer UI in Candy DC for even more effective writing;
- Resource folders (including custom paths);

Have a look in the subfolders in **Candy_DE/Documents/Guides** to find what you need!